19 $y = \dfrac{d^2}{m}$

d = 11.3 correct to 3 significant figures.
m = 8.4 correct to 2 significant figures.

By considering bounds, calculate the value of y to a suitable degree of accuracy.
You must show all your working and give a reason for your final answer.

$11.25 \leq d < 11.35;\ 8.35 \leq m < 8.45$

1 mark
Both measurements round to nearest0.1.

Lower y bound = $11.25^2 \div 8.45 = 14.97781065...$

1 mark
Lower numerator; upper denominator.

Upper y bound = $11.35^2 \div 8.35 = 15.42784431...$

1 mark
Upper numerator; lower denominator.

Both bounds round to15 to the nearest whole number.

They do not both round to the same 1 d.p. value.

Therefore y = 15 to the nearest whole number.

1 mark
Choose the greatest accuracy guaranteed by BOTH bounds.

(Total for Question 19 is 4 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA D